**Federal and State Tax Parameter Workbook Documentation**[1]

The U.S. Census Bureau imputes federal, state, local, and payroll tax liabilities and tax credits via a microsimulation tax model using the Current Population Survey Annual Social and Economic Supplement (CPS ASEC), the Internal Revenue Service (IRS) Statistics of Income Public Use File (SOI PUF), and information from federal and state tax forms and instructions.[2] The microsimulation model contains two primary components: a collection of SAS programs and a workbook of federal and state tax parameters.

The SAS programs calculate payroll taxes; create tax units; perform a statistical match with the IRS SOI PUF; calculate federal, state, and select local income taxes; and output key tax variables.[3]

The federal and state parameter workbook holds the key parameters used in the calculation of federal, state, and local income taxes for the 2023 – 2025 CPS ASEC tax models.[4] The information contained in the parameters include standard deduction amounts, tax brackets and rates, and thresholds and amounts used to calculate tax credits, among others. In addition, the workbook describes where the parameters can be found in tax forms and/or instructions, making it easier to reference additional information on the parameter and to update the parameter each year.

In the workbook, there is a spreadsheet with all the parameters needed to calculate federal payroll and income tax, plus spreadsheets with the parameters for each state with an income tax.[5] Each of the spreadsheets follows the same structure. The first column in each spreadsheet ("Parameter") includes the variable name used for the parameter in the tax model SAS programs. The second column ("Description") provides a brief description of what is captured by the parameter. The subsequent columns include the parameter values and sources for each year in the workbook. For each year, the first column ("[Year] Value") contains the numerical parameter or array for that tax year.[6] The second column ("[Year] Source") describes where the value can be found in federal or state tax forms and/or instructions.[7]

There are limitations to the information captured in the parameter workbook.

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[1] Suggested citation: Shantz, K. (2026). *Federal and State Tax Parameter Workbook Documentation: 2023-2025*. U.S. Census Bureau. https://www2.census.gov/library/working-papers/2026/demo/Federal_and_State_Tax_Parameter_Workbook_Documentation_TY2025.pdf.

[2] The 2023 – 2025 tax models use the 2015 IRS SOI PUF file in the statistical match.

[3] This release does not include the SAS programs.

[4] The parameters for tax years 2023 – 2025 were those used for the 2023 – 2025 CPS ASEC tax models to generate the federal and state tax liability and credits for the 2024 – 2026 CPS ASEC files.

[5] As of tax year 2023 and 2024, there is no state income tax in Alaska, Florida, Nevada, South Dakota, Tennessee, Texas, Washington, and Wyoming. As of tax year 2025, there is no longer state income tax in New Hampshire. The parameter workbook includes the state tax parameters for New Hampshire for tax years 2023 and 2024. Washington has a state earned income tax credit and is included in the parameter workbook as of tax year 2024. Other than New Hampshire and Washington, states without state income taxes are not included in the parameter workbook.

[6] For some parameters, we capture a single numerical value, such as the federal standard deduction for single filers. For other parameters, we capture an array with multiple values, such as the federal taxable income brackets to determine the marginal tax rate for a single filer.

[7] Tax forms and instructions can generally be found on federal or state websites for their department of revenue or taxation. The source sheet in the workbook contains links to the federal and state websites with forms and instructions (last checked in February 2026).

- The federal and state parameters are not updated or adjusted retroactively. The parameters for each tax year captured in this release were used to generate the federal and state tax liability and credits in that year's Supplemental Poverty Measure (SPM) and CPS ASEC microdata.
- In the tax model, we capture federal and state policies for the following filing statuses: single, head of household, and married filing jointly. We do not capture information on married filing separately or qualified widow(er), even in states in which it may be more advantageous for married couples to file separately.
- Additionally, not all federal and state tax policies are captured in the tax model. We do not include policies that we do not have data to model from the CPS ASEC or the IRS SOI PUF. The policies not modeled include use tax, circuit breaker credits, and renter credits. In some cases, an aspect of state tax policy is modeled with simplifying assumptions because information is not available in the data.[8]
- The tax model captures local income taxes in some, but not all, states with local income taxes. The model calculates local income tax liabilities and credits in four states: Indiana, Iowa, Maryland, and New York. [9]

We welcome the comments and questions of data users. Additionally, feel free to reach out to request a 508 compliant version of the document or other accommodation. For comments, questions, and requests, please email: sehsd.isb.list@census.gov.

[8] For example, in Maine, the amount of the child care credit depends, in part, on the quality rating of the child care facility the child attends. Because we do not have information on the quality rating, we assume all filers' children are receiving care in facilities that are not designated as high quality (designated as Step 4 or Step 5 by the state).
[9] We include local income taxes that use a state definition of income and are administered and collected by the state. In these states, residents file and pay their local income tax using their state income tax return. There are an additional 12 states with local income tax (as of February 2023). Refer to the CPS ASEC Tax Model User Notes for Tax Year 2023 for more information.